1. $L_1: \dfrac{x-1}{1} = \dfrac{y-2}{0} = \dfrac{z-3}{-1}, L_2: \dfrac{x+2}{2} = \dfrac{y-1}{1} = \dfrac{z}{1}.$

Find the plane passing through L_1 and parallel to L_2.

2. Does the series $\displaystyle\sum_{n=2}^{\infty}(-1)^{n+1}\frac{\ln^2 n}{n^p}$ converge or diverge?

3. Find convergence set and sum for the power series $\displaystyle\sum_{n=1}^{\infty}(-1)^{n-1}\left(1+\frac{1}{n(2n-1)}\right)x^{2n}.$

4. Expand the function $f(x) = \displaystyle\sum_{n=1}^{\infty}\left(\frac{x}{1-x}\right)^n$ into power series of x.

5. Determine the continuity of $f(x,y) = \begin{cases} \dfrac{x^2 \sin\dfrac{1}{x^2+y^2} + y^2}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$

6. $f(x,y) = \begin{cases} xy\,\dfrac{x^2-y^2}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$, find $f_{xy}(0,0), f_{yx}(0,0)$.

7. Let $z = f\left(xy, \dfrac{x}{y}\right) + g\left(\dfrac{y}{x}\right)$, f has a second − order continuous partial derivative,

g has a second − order continuous derivative, find $\frac{\partial^2 z}{\partial x \partial y}$.

8. Find the tangent line to $\begin{cases} x^2+y^2+z^2 = 6 \\ z = x^2+y^2 \end{cases}$ at $(-1,1,2)$.

9. Let $u = f(x,y,z)$ has a continuous partial derivative, $z = z(x,y)$ is determined by $xe^x - ye^y = ze^z$, find du.

10. Find the maximum and minimum values of $z = f(x,y) = x^2y(4-x-y)$ on the region D, $D = \{(x,y)|x \geq 0, y \geq 0, x+y \leq 6\}$.

11. Find the minimum distance from $y = x^2$ to $x - y - 2 = 0$.

12. Find $\displaystyle\iint_D \frac{3x}{y^2+xy^3}dxdy$, D is bounded by $xy = 1, xy = 3, y^2 = x$ and $y^2 = 3x$.

13. Compute the triple integral $\displaystyle\iiint_\Omega x^2\sqrt{x^2+y^2}\,dxdydz,$

Ω *is* bounded by $z = \sqrt{x^2+y^2}$ and $z = x^2+y^2$.

14. Let L be $y = \pi\cos x$ from $A(\pi,-\pi)$ to $B\,(-\pi,-\pi)$, find $\displaystyle\int_L \frac{(x+y)dx-(x-y)dy}{x^2+y^2}$.

15. Find $I = \displaystyle\iint_\Sigma \frac{1}{\sqrt{x^2+y^2+(z-a)^2}}dS$, $\Sigma: x^2+y^2+z^2 = R^2, 0 \leq a < +\infty, a \neq R.$

16. Find $\displaystyle\iint_\Sigma \frac{axdydz+(z+a)^2dxdy}{\sqrt{x^2+y^2+z^2}}$, $\Sigma: z = -\sqrt{a^2-x^2-y^2}$, $(a > 0)$

the direction is upper side.

17. Find the solution of the equation $(2x+y-4)dx+(x+y-1)dy = 0$.

18. Find the solution of the equation $y'' - 2y' + 2y = e^x\cos x + e^{2x}$.